Title of Abstract **Off the Grid Hydrogen Production**

Name of Author **Ass. Prof. Anna-Karin Axelsson**

Name of University London South Bank University

Affiliation

London South Bank University, Centre of Advanced Materials, School of Engineering, 103 Borough Road, SE1 0AA London, United Kingdom

Abstract

Hydrogen production solutions with reduced environmental impact have attracted increased attentions. Majority of global hydrogen production is currently undertaken by steam reforming of fossil fuels which is a highly energy demanding process in itself

The lecture will present different solutions of producing hydrogen off the national grip which can support rural areas and developing countries to expand the usage of hydrogen fuel cells as a power solution.

The lecture will focus on the research undertaken at London South Bank University such a water-metal particle hydrogen generation and thermochemical cycles for hydrogen generation.

Thermochemical cycles can be employed to produce both oxygen and hydrogen for fuel cell power generation. The fluid cycle is separated into two distinct parts where heat sources (thermo side) produce oxygen and the electrochemical (chemical side) produce hydrogen. Despite hydrogen producing thermochemical cycles have been invested the last forty years, not much evidence is found to make these cycles completely renewable (or clean technology) energy driven. Therefore, this research present a hybrid solar thermochemical cycle solution that can be developed utilizing the complete solar spectrum (both the quantum and thermal part) producing a hydrogen generation solution working off the electrical grid.

It will show how united engineering effort can take future hydrogen production to a cleaner and greener process alternative.

Biography

Dr Anna-Karin Axelsson is currently Associate Professor in Functional Materials in the Centre of Advanced Materials at London South Bank University, and a Visiting Fellow at Department of Materials, Imperial College London.

She is also highly engaged in academic teaching excellence as Course Director of postgraduate courses in Chemical and Process Engineering Management and undergraduate courses in Chemical and Process Engineering within School of Engineering.

Dr Axelsson have several years of both industrial- and education experiences within the field of Mechanical- and Chemical Engineering and since year 2000 been mainly engaged in the field of Materials Science in academia.

She has a strong track record in successful collaboration with other research teams, institutions and industry with track record of significant and sustained success in obtaining research grants

She has successfully supervised 6 PhD students and 10 MPhil to completion during her years at Imperial College London and London South Bank University, presented at 70 International Conferences worldwide and as a result, Dr Axelsson have been invited to numerous Conference Committees and presented several times as a prestigious Plenary and Keynote Speaker.

Anna Axelsson has a H-index of 22 and total of 1453 citation and over 60 peer-reviewed articles

Photo

Email id: a.k.axelsson@lsbu.ac.uk